MIL-F-52443D
20 October 1975
SUPERSEDING
MIL-F-52443C
15 April 1970

MILITARY SPECIFICATION

FILTER, INFRARED INTERFERENCE-ABSORPTION TYPE, FOR SEARCHLIGHT, INFRARED, 28-VOLT, 110-AMPERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of interference-absorption infrared filter. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping and Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing

FSC 5855

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PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive Filament Reinforced

MILITARY

MIL-P-116 MIL-P-11268	Preservation Packaging Methods of Parts, Materials, and Processes Used in Electronic Equipment
MIL-B-43014	Boxes: Water Resistant Paperboard Folding, Set-up, and Metal-Stayed

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets Skids, Runners, or Pallet Type Base
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-D-613921	Filter, Infrared, Interference-Absorption Type

(Copies of specifications, standards, and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description. The filter, infrared, interference-absorption type, hereinafter referred to as the filter, shall be in accordance with Drawing SC-D-613921 and as specified herein.

3.2 Construction. The filter shall be in accordance with Drawing SC-D-613921 and all provisions of this specification.

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3.3 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Operational and environmental.

3.5.1 Mechanical shock. The filter shall not be damaged (see 6.3.1) by impacts of 3 foot pounds.

3.5.2 Interference-coating. The interference-coating shall be applied on the concave surface over the usable zone of the filter.

3.5.3 Interference-coating bond. The interference-coating bond shall prevent the separation of the interference-coating from the absorption filter.

3.5.4 Interference-coating radiation exposure. The filter shall withstand, without damage, direct radiation exposure that will result in a stabilized temperature, measured at the apex of the convex side of the filter, of plus 204°C.

3.5.5 Spectral transmission. Spectral transmission of the usable and nonusable zones of the filter shall be as follows:

a. Usable zone. Spectral transmission of the usable zone shall be between the minimum and maximum values shown on Figure 1 for any wavelength from 800 to 1,100 millimicrons.

b. Nonusable zone. Spectral transmission of the nonusable zone shall be between the minimum and maximum values shown on Figure 2 for any wavelength from 400 to 1,100 millimicrons.

3.5.6 Visual security. There shall be no evidence of visible white light at any distance greater than 25 feet.

3.5.7 Storage temperature.

3.5.7.1 Low temperature storage. The filter shall not be damaged by storage in an ambient temperature of minus 62°C.

3.5.7.2 High temperature storage. The filter shall not be damaged by storage in an ambient temperature of plus 68°C with humidity controlled at 90 to 98 percent.

3.5.8 Thermal shock. The filter shall not be damaged by thermal shock from minus 54°C to plus 260°C.

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3.6 Identification marking. The filter shall be marked and identified in accordance with MIL-STD-130.

3.7 Workmanship. The filter shall show no evidence of chips, cracks, striations, or any imperfection which could impair the normal functions of the filter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified by the contract or purchase order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements.

4.1.1 Parts and materials inspection. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

c. Inspection of preparation for delivery (see 4.8).

4.3 First article inspection.

4.3.1 Inspection. Each filter shall be examined in accordance with Table I. One or more defects shall be cause for rejection of that filter.

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TABLE I Inspection

Defects	Requirement paragraph
Major	
101. Dimensions not as specified	3.2
102. Parts and Materials not as specified	3.4
103. Absence of interference coating	3.5.2
104. Marking not in accordance with MIL-STD-130	3.6
105. Workmanship not as specified	3.7
Minor	
Defects not listed as major shall be classified as minor.	

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, the first article filters shall be subjected to the inspections in Table II. Inspections may be conducted in any sequence. Failure of any test shall be cause for rejection.

TABLE II First article inspection

Inspection	Requirement paragraph	Test paragraph
Mechanical shock	3.5.1	4.6.1
Interference-coating	3.5.2	4.6.2
Interference-coating bond	3.5.3	4.6.3
Interference-coating radiation exposure	3.5.4	4.6.4
Spectral transmission	3.5.5	4.6.5
Visual security	3.5.6	4.6.6
Low temperature storage	3.5.7.1	4.6.7.1
High temperature storage	3.5.7.2	4.6.7.2
Thermal shock	3.5.8	4.6.8

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4.4 Inspection covered by subsidiary documents: The filter shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order (see 4.3.1, Table I).

4.5 Quality conformance inspection.

4.5.1 Inspection. Each filter shall be inspected as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests.

4.5.2.1 Group A Inspection. Each filter shall be inspected in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Mechanical shock	3.5.1	4.6.1
Interference coating	3.5.2	4.6.2
Interference coating bond	3.5.3	4.6.3

4.5.2.2 Group B Inspection. Group B inspections shall be conducted on filters selected from units which have passed the examination and tests in 4.5.1 and 4.5.2.1. Sampling shall be in accordance with MIL-STD-105, AQL 6.5 percent defective, inspection level S-2. Tests shall be conducted in accordance with Table IV and may be performed in any sequence.

TABLE IV Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Spectral transmission	3.5.5	4.6.5
Thermal shock	3.5.8	4.6.8

4.5.2.3 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspections. Sample units shall be selected in accordance with 4.5.2.3.1.

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4.5.2.3.1 Sampling for inspection. Three (3) sample filters shall be selected at random for all inspections in Table V from the monthly specified quantity (see 6.2) produced or each one-hundred (100) filters whichever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE V Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Interference-coating radiation exposure	3.5.4	4.6.4
Visual security	3.5.6	4.6.6
Low temperature storage	3.5.7.1	4.6.7.1
High temperature storage	3.5.7.2	4.6.7.2

4.5.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C reinspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test methods. Tests required to determine compliance with this specification shall be conducted in accordance with the test procedures specified herein. All test gages and instruments used during tests shall have been calibrated within 30 days prior to the start of testing and thereafter as required, or at intervals not to exceed 6 months. Unless otherwise specified all tests shall be performed at room temperature (see 6.3.2). The spectrophotometer shall be calibrated to read the transmission within plus or minus 3 percent of full scale for any wavelength. The spectrophotometer shall be capable of measuring transmission from 100 percent to 0.01 percent from 400 millimicrons to 1,100 millimicrons. Test facilities, chambers, and apparatus used in conducting tests shall meet the requirements of MIL-STD-810.

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4.6.1 Mechanical shock. The filter shall be mounted on a board larger than the filter and shall be covered with a hard paper between 0.012 and 0.020 inch thick during the test. A steel ball weighing 3 pounds shall be dropped three times on the filter from a height of 1 foot. Any evidence of damage to the filter shall constitute failure of this test.

4.6.2 Interference-coating. The interference-coating shall be examined in accordance with Table I. Any evidence of visual defects in the coating shall be cause for rejection.

4.6.3 Interference-coating bond. A strip of clear 3/4 inch wide cellophane tape approximately 10 inches long shall be applied to the interference-coating across the major arc and allowed to set for 5 minutes. At the end of 5 minutes, a 290 gram weight shall be attached to the free end of the tape and the weight shall be allowed to hang free of the interference-coating surface at an angle of 90° from the concave surface for a period of 5 minutes. Then the tape shall be pulled off the coating at approximately a 90° angle in less than 1 second. Any evidence of interference-coating on the tape after removal from the filter shall constitute failure of this test.

4.6.4 Interference-coating radiation exposure. The filter shall be mounted in a test fixture approved by the Government, which will transmit a direct radiation energy through from the concave side of the filter to the convex side of the filter, which will result in an equilibrium temperature at the apex of the convex side of the filter of plus 204°C. The convex side of the filter shall be exposed to room temperature during this test. The filter shall be exposed for a period of not less than 20 hours after an equilibrium temperature of plus 204°C has been reached. At the end of 20 hours, the filter shall be examined for damage. Any evidence of damage shall constitute failure of this test.

4.6.5 Spectral transmission. The spectral transmission of the filter shall be measured at 0°, 30°, 40°, 60°, and 75° from the center of the filter with a spectrophotometer. These measurements are to be made after the filter has reached thermal equilibrium of plus 204°C in an approved testing device. The spectral transmission curves for each position measured on the filter shall be plotted on one sheet of 5 cycle semi-logarithmic graph paper. Failure to meet the spectral transmission requirement of 3.5.5 shall constitute failure of this test.

4.6.6 Visual security. The filter shall be mounted in a test fixture simulating a searchlight, infrared, 28-volt dc, 110-ampere. The test fixture shall offer no obscuration between the light source and the filter. The light source shall be operated in an ambient light not to exceed 2×10^{-5} footcandle under normal operating current and voltage. Three observers shall be dark adapted for at least 30 minutes while the filter is temperature stabilizing. The observers shall have normal or

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corrected 20/20 vision and shall have no defects of vision which would affect the validity of the test. After the dark-adaptation period, each observer shall look at the filter up to 90° from center. Each observer shall be positioned 25 feet from the filter. Any white light visible to two of the observers shall constitute failure of this test.

4.6.7 Storage temperature.

4.6.7.1 Low temperature storage. The filter shall be stored at an ambient temperature of minus 62°C for 4 hours. Any evidence of damage shall constitute failure of this test.

4.6.7.2 High temperature storage. The filter shall be stored at an ambient temperature of plus 68°C for 4 hours with humidity controlled at 90 to 98 percent. Any evidence of damage shall constitute failure of this test.

4.6.8 Thermal shock. The filter shall be exposed to an ambient temperature of minus 54°C for not less than 2 hours. The filter shall then be exposed to plus 260°C within 2 minutes, for not less than 2 hours. This constitutes one cycle. The filter shall be exposed to 4 cycles of thermal shock. Any evidence of damage shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each filter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each filter shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation applications. None required.

5.1.1.4 Unit packaging. Each filter shall be individually packaged, Method III in accordance with MIL-P-116 as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, type I and secure wrap with tape conforming to PPP-T-45, Type III, grade A. Place the cushioned item within a close-fitting box conforming to MIL-B-43014, form 1, style II, type A. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.1 Intermediate container. Filters packaged as specified in 5.1.1.4, shall be placed in intermediate container conforming to MIL-B-43014, form 1, style II, type A in multiples of five, not to

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exceed 50 unit packages or 20 pounds. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C Filters shall be preserved and packaged in a manner that will afford adequate protection against physical and enviornmental damage during shipment, handling and limited in-transit storage.

5.2 Packing Packing shall be Level A, B, or C as specified. Shippin containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A.

5.2.1.1 Consolidation. A quantity of filters packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicabl

5.2.1.2 Palletized load. A quantity of containers, packed as specifie in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147, except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accomodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packe within a close fitting box conforming to PPP-B-621, overseas type; PP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceed 100 pounds, 3 x 4 inch skid laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B.

5.2.2.1 Consolidation. A quantity of filters packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

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5.2.2.2 Palletized load. A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. A quantity of filters packaged as specified in 5.1 shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The interference filter is intended for use in a searchlight infrared conforming to MIL-S-52444.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and schedule for first article testing for all line items (see 4.3).

c. Production delivery schedule (see 4.5.2.3.1).

d. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

e. Disposition of group C failures.

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6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Damage. Damage is defined as chipping, breaking, pitting, cracking, or peeling.

6.3.2 Room temperature. Room temperature is defined as plus 23°C, plus or minus 3°C.

CUSTODIAN:

Army-EL
Navy-EC
Air Force-84

REVIEW ACTIVITY:

Army-AT
Navy-OS

PREPARING ACTIVITY:

Army-EL

Project No. 5855-000

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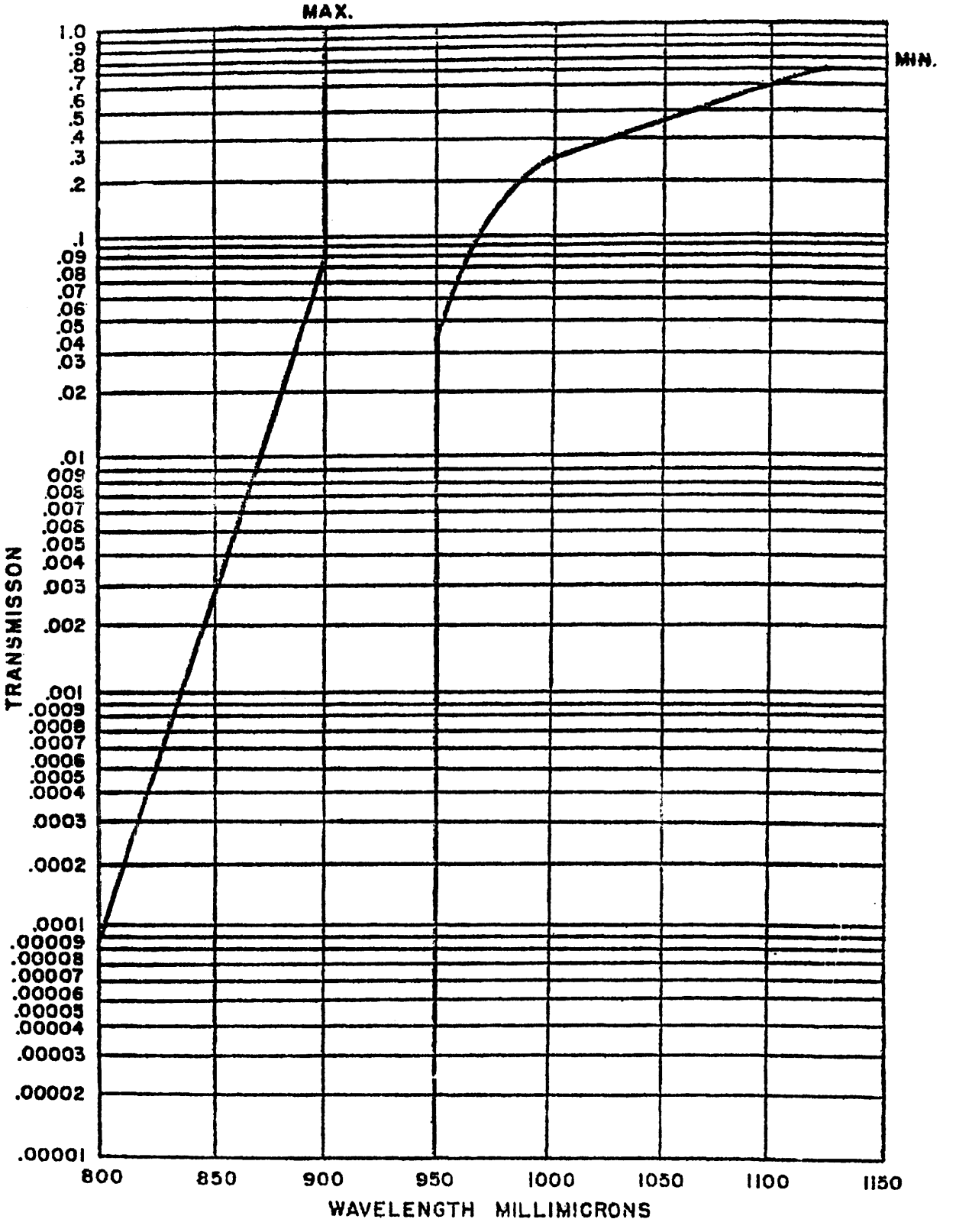


FIG. 1 SPECTRAL TRANSMISSION OF USABLE ZONE OF FILTER

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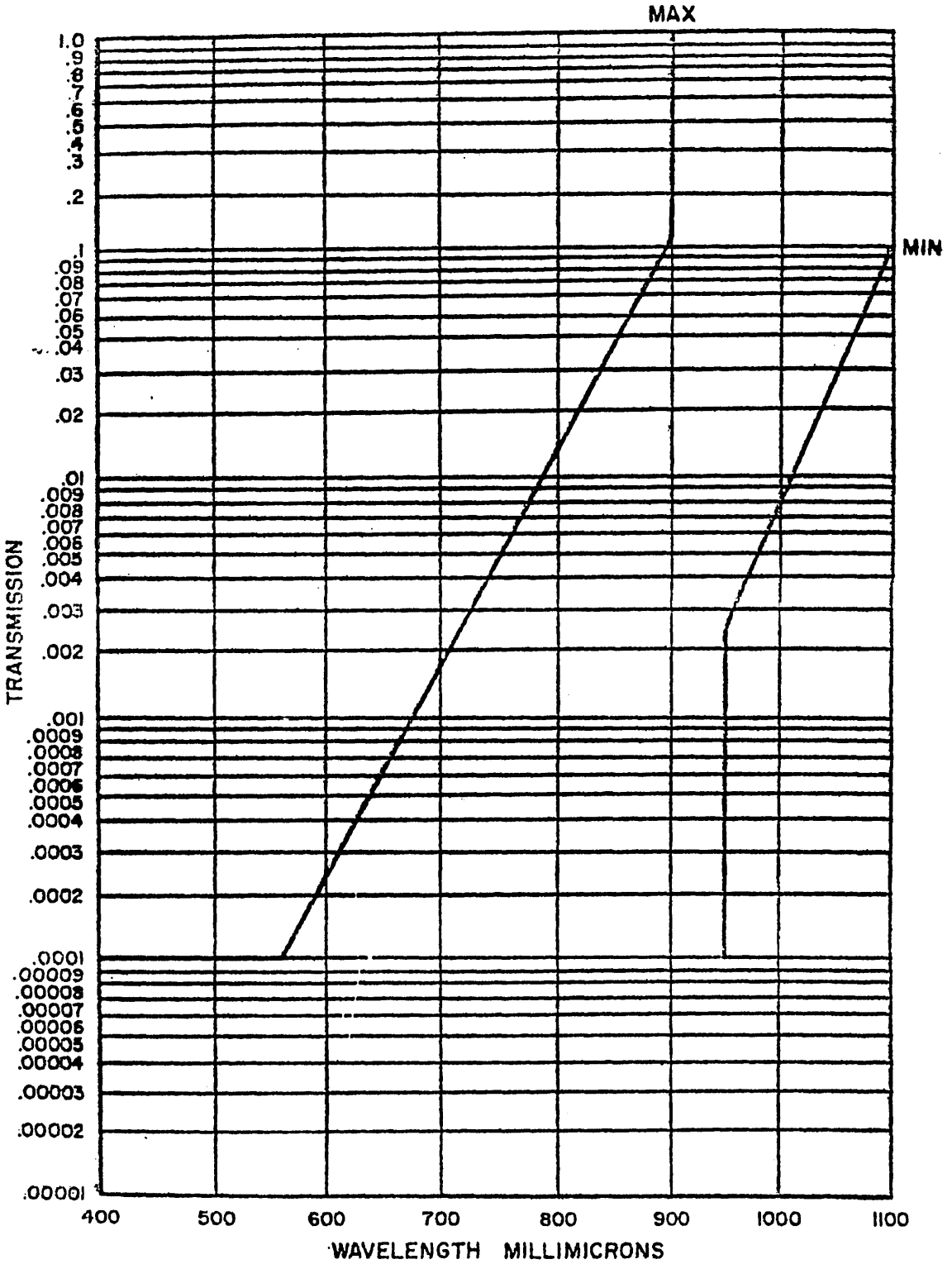


FIG. 2 SPECTRAL TRANSMISSION OF NONUSABLE ZONE OF FILTER

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-F-52443D FILTER, INFRARED INTERFERENCE-ABSORPTION TYPE, FOR SEARCHLIGHT, INFRARED, 28 VOLT, 110- AMPERE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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